

Project (40%)

Subject Course: CDS6214 Data Science Fundamentals

Semester: 2610

Submission Deadline: 28th June 2026 (Sunday), 11:59pm

1. Overview & Objectives

This project requires students to apply the full data science workflow on a real-world dataset using Python and Jupyter Notebook. Upon successful completion of the project, students will be able to execute a complete data analysis pipeline, including selecting an appropriate dataset, applying suitable analytical and visualization techniques, and generating useful, evidence-based insights that clearly address the defined problem statement and support effective, data-driven decision-making.

You will:

- Select and justify a dataset within your assigned domain (with tutor approval).
- Produce a literature review to understand the context of your topic.
- Construct a clear problem statement supported by four diverse analytical questions.
- Implement a reproducible pipeline involving data loading, cleaning, EDA, visualization, models building and performance evaluation.
- Present key insights through both the notebook and a recorded presentation.

2. Group Formation

Group Size

Form a project group of up to four (4) students. All members must belong to the same tutorial section.

Group Registration Procedure

Consult your tutor for instructions on forming and registering your group. Group membership is considered final only after tutor approval.

3. Domain Allocation

Each tutorial session is assigned specific domains to ensure variety across projects. You must select a dataset that clearly fits within your tutorial's assigned domain.

Tutorial section	Domains
TT1L	Healthcare & Biomedical Data
TT2L	Finance, Economics & Business Analytics
TT3L	Environment, Climate & Sustainability
TT4L	Social Sciences & Demographics
TT5L	Transportation, Mobility & Urban Analytics
TT6L	Sports, Entertainment & Human Performance

Your dataset must comply with the Dataset Requirements section and must be approved by your tutor.

4. Deliverables

- Jupyter Notebook (TTXL_GXX_Notebook.ipynb) with README (first cell).
- Dataset download link (original source page) in README.
- YouTube link to recorded presentation video in README.
- Presentation slides (TTXL_GXX_Presentation.pptx).

5. Dataset Requirements

- Must come from a recognized open data source (e.g., Kaggle, Google Dataset Search, Data.gov, UCI Machine Learning Repository, government open data portals, reputable research repositories).
- Provide a public, direct link to the original dataset page (not personal drives).
- Do not submit modified data files; the examiner will download the original dataset from your link.
- Include dataset citation and license.
- Check this Dataset Requirements section carefully before selecting your dataset.

Tutor approval is required before proceeding.

6. Project Requirements

A. Domain Understanding (Literature Review)

Provide a brief summary of the domain, relevant work, and context.

B. Data Science Pipeline

Follow the structure provided in TTXL_GXX_Notebook.ipynb: data loading → cleaning/preprocessing → EDA → analysis/modeling → evaluation → insights → conclusions → references.

C. Problem Statement & FOUR Analytical Questions

Write a clear problem statement. Develop four analytical questions that collectively support solving that problem. Each question must use a distinct analytical technique or algorithm; questions with overlapping methods will be counted as one analysis.

D. Insight Generation

Conduct analysis and/or modeling to derive meaningful insights that directly support the problem statement and questions.

E. Dataset Loading Rule (Strict)

Load your dataset once only. If additional copies are needed, create them programmatically from the initial in-memory dataset. Any cell that reloads the dataset from file will be ignored and receive zero marks.

7. Technical & Reproducibility Requirements

- Python-only workflow in Jupyter Notebook.
- Use relative paths only (e.g., ./data/).
- Avoid hidden manual steps; all transformations must be in code.
- Fix random seeds for reproducibility.
- Print Python and library versions; include requirements.txt/environment.yml where possible.
- Avoid internet dependencies at runtime; if necessary and permitted, document clearly.
- Notebook must run top-to-bottom without errors on the examiner's machine.

8. Presentation (PPTX and Video)

- Prepare a PowerPoint presentation (about 10 slides).
- Record a group presentation (maximum 10 minutes) and upload it to YouTube as an unlisted video.

- All group members must present and appear on camera.
- Penalty applies on poor audio and/or screen recording quality (e.g., inaudible audio, unreadable text/figures).
- Include the YouTube link in the README section of your Jupyter Notebook. Please note that the examiner will not notify you if your video link is unavailable, inaccessible, or incorrectly provided. It is the responsibility of the group to ensure that the video link is fully functional and accessible at the time of assessment. Failure to provide a valid and working link will result in a score of zero (0) marks for the presentation component.
- Be prepared for a possible in-person presentation if requested by the examiner.

9. Presentation Content:

In the presentation, students should provide background context for the problem and introduce the dataset clearly. This includes explaining what the dataset contains, why it was selected, and how the data relates to the problem being studied. Students must also describe the real-world relevance of the dataset—how it is used or how it could be applied in practical, real-life scenarios. The presentation must include:

- Introduction (including project background, dataset introduction, and real-life relevance).
- Problem Statement and analytic questions
- Data analysis and visualizations
- Insights and application
- Conclusion

10. File Naming Convention

Name all your files according to your tutorial session and group ID (e.g., TT1L_G01):

- TT1L_G01_Notebook.ipynb
- TT1L_G01_Presentation.pptx
- TT1L_G01_Requirements.txt (if provided)

Failure to follow the naming convention will incur penalties (see Penalties Section).

11. Submission Instructions

- Submit all required components via EbWise before the deadline.
- Late submissions incur a 10-mark deduction per hour.
- Submissions more than 4 hours late will receive zero marks.
- Do NOT resubmit after the deadline; penalties apply.
- Submission via personal/student drive/email is not a valid submission and will not be graded.

- DO NOT submit zipped files.

12. Penalties

Violation	Penalty
Late submission (per hour)	–10 marks
Submission >4 hours late	0 marks
Plagiarism	0 marks
File not following naming convention	–5 marks per file
Video exceeds 10 minutes	–5 marks
Video link inaccessible	–8 marks
Missing Jupyter Notebook file	0 marks
Missing power point file	–8 marks
Missing information in README	–20 marks
Using data from an unassigned domain	–20 marks
Using data unapproved by tutor	–20 marks

13. Academic Integrity Reminder

All submitted work must be your group’s own original work. Plagiarism—including uncredited copying of text, code, figures, or results—will result in an immediate award of 0 marks for the project and may trigger further disciplinary action. Cite all external sources, including datasets, articles, and any code snippets used for reference.

14. Final Self-Checklist

- Dataset link is valid and from a recognized source; tutor approval obtained.
- Dataset loaded once only; relative paths used throughout.
- Four analytical questions use distinct methods; no overlapping counted twice.
- Notebook runs end-to-end without errors on an external machine.
- File naming convention followed (e.g., TT1L_G01_*).
- All deliverables submitted to eBwise on time, necessary links included.

Project Rubrics (40%)

Code : Data Science Pipeline (20%)

Criterion	4 Marks (Excellent)	3 Marks (Good)	2 Marks (Satisfactory)	1 Mark (Weak)	0 Marks (Missing)
1. Problem Definition & Research Questions	Clear, focused objectives; well-formulated research questions fully aligned with dataset. Dataset from recognized source.	Mostly clear objectives; good alignment with dataset. Dataset valid.	Objectives somewhat vague; partial alignment. Dataset acceptable but documentation incomplete.	Objectives unclear; research questions poorly defined. Dataset source unclear or poorly documented.	No meaningful objectives or research questions. No dataset link or invalid/unverifiable dataset.
2. Data Cleaning & Exploratory Data Analysis (EDA)	Thorough cleaning with clear justification; correct handling of missing values, types, and outliers. Insightful EDA; strong visualizations; interpretations clearly explained.	Good cleaning, adequate EDA with relevant visuals; some interpretation missing.	Basic cleaning; Basic EDA with limited visual analysis, important steps missing.	Minimal or incorrect preprocessing, very minimal EDA; little/no interpretation.	No data cleaning and/or EDA performed.
3. Visualization Quality	Visualizations are clear and well-designed; they effectively support	Visualizations are generally clear and appropriate; minor issues in labeling,	Visualizations are basic or inconsistently formatted;	Visualizations are unclear, poorly chosen, or incorrectly labeled;	No meaningful visualizations included, or visual content is entirely

	analytical reasoning; charts are correctly chosen, labeled, and interpreted; figures enhance understanding and narrative flow.	formatting, or chart choice; figures support analysis and communication but may lack refinement or full interpretive depth.	occasional inaccuracies or poor chart choices; labeling may be incomplete; figures provide some insight but do not fully aid communication.	charts fail to communicate insights; figures hinder rather than support understanding; weak integration with narrative.	irrelevant or incorrect.
4. Analysis / Modeling	Appropriate methodology; well-explained; metrics correct; insightful interpretation.	Mostly appropriate method; limited explanation.	Weak or basic analysis; limited metrics.	Poor method selection; results unclear.	No analysis or modeling performed.
5. Reproducibility & Code Execution	Notebook runs end-to-end without errors; relative paths; version details included; works with original dataset.	Mostly reproducible; minor fixes needed.	Partial reproducibility; several execution issues.	Significant errors; code difficult to run.	Notebook fails to run with original dataset.

Non-negotiable: If the notebook fails to execute on the examiner's side with the original dataset from your link, the Code and Presentation components will be graded as failed.

Presentation (20%)

Criterion	4 Marks (Excellent)	3 Marks (Good)	2 Marks (Satisfactory)	1 Mark (Weak)	0 Marks (Missing)
6. The FOUR Questions (Distinct Analysis Techniques, refer to Appendix)	All four questions use distinct and correctly applied analysis techniques/algorithms; no overlap in methodological approach.	Three questions are distinct; one partially overlaps in technique.	Two questions use distinct methods; two overlaps significantly.	Only one distinct analysis; most overlap or reuse the same technique.	No question attempted.
7. Results, Insights & Interpretation	Strong insights backed by evidence; clearly linked to research questions.	Good insights; minor gaps in explanation.	Limited or generic insights; weak connection to questions.	Very weak interpretation; results not explained.	No results or insights presented.
8. Delivery & Communication Skills	Clear, confident delivery; strong voice projection; excellent pacing; highly engaging; minimal filler words.	Generally clear and confident; pacing mostly appropriate; minor lapses.	Understandable delivery but inconsistent clarity or pacing; some read from slide; limited engagement, some read from slides	Weak delivery; difficult to follow; mostly read from slide; frequent filler words, mostly read from slides	No meaningful delivery; presenter inaudible, unclear, or did not present.
9. Understanding & Explanation	Demonstrates excellent and in-depth understanding; concepts and insights explained clearly and accurately.	Good understanding with mostly clear explanations; minor inaccuracies or gaps.	Basic understanding; explanations sometimes unclear or incomplete.	Limited understanding; frequent inaccuracies; explanations confusing.	No demonstration of understanding; unable to

					explain content.
10. Presentation Slides Content	Slides have an excellent balance of figures and text; content flows logically with strong visual storytelling.	Slides maintain a reasonable balance between text and figures; overall flow is mostly coherent.	Slides contain noticeable text-heaviness or insufficient visuals; flow is uneven.	Slides contain excessive text with minimal or irrelevant figures; flow is difficult to follow.	Slides provided, or visuals/slides do not relate to the project.

Appendix:

What counts as a distinct analysis technique?

A technique is considered distinct if it:

- Has a different analytical goal (e.g. prediction, pattern discovery, explanation),
- Uses a different underlying method or algorithm, and
- Produces a different type of result (e.g. predictions, clusters, statistical tests).

Changing parameters, datasets, or evaluation metrics does not count as a new technique.

Examples of Non-Distinct Techniques

Example	Why It Is NOT Distinct
Logistic regression vs. linear regression	Same model family; different output scale
K-means on Dataset A vs. Dataset B	Same technique, different data
Decision tree with different depths	Parameter tuning, not methodological change
Cosine similarity vs. Euclidean distance in clustering	Distance metric variation only
Accuracy vs. F1-score	Evaluation metrics, not analysis techniques